

AVANISH GRAMPUROHIT

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EDUCATION

Pennsylvania State University

Bachelor of Science in Computer Science | GPA: 3.4/4

University Park, PA

Aug 2021 – May 2025

- **Coursework:** Machine Learning, Neural Networks, Vision and Language, Software Development, Computer Vision, Data Structures and Algorithms, Database Management Systems, Operating Systems

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++
- **Web Development:** React.js, Node.js, RESTful API
- **Data & AI:** PyTorch, Sci-kit Learn, HuggingFace, NumPy, Pandas, Matplotlib, MongoDB, MySQL
- **Miscellaneous:** OpenCV, Conda, Keras, Linux, AWS Cloud(S3, EC2), Git (Version Control)
- **Concepts:** Action Recognition, Large Language Models (LLMs) Object Tracking, Image Compression, Transformers, Natural Language Processing (NLP)

EXPERIENCE

Deep Learning Researcher

Pennsylvania State University | Dr.Huijuan Xu

University Park, PA

May 2024 – May 2025

- Pioneered action recognition research, analyzing the effectiveness of **5+ state-of-the-art (SOTA) algorithms** over a decade.
- Collected and processed **500+ hours of video** from **10+ large-scale instructional video datasets**
- Streamlined data extraction with **asynchronous functions**, using **BeautifulSoup4** to scrape YouTube and extract data, reducing processing time by **85%**.
- Designed and executed experiments for **5 SOTA deep learning models** (e.g., **InternVideo2, ViViT, I3D, R3D, VideoSwin**) to test action recognition.
- *Tools: Python, PyTorch, OpenCV, NumPy, Pandas, Matplotlib*

Research Assistant

Pennsylvania State University | Dr.Mahfuza Farooque

University Park, PA

Aug 2023 – May 2024

- Investigated social media's impact on depression, analyzing **100M+** posts and **100** manually annotated entries from 4 researchers.
- Engineered an **open-closed vocabulary pipeline** using **Hugging Face** and **BERT**, achieving **86%** accuracy in sentiment prediction across 3 datasets.
- Extracted **6 features** from data, leveraging insights from **10+ psychology studies** to improve model performance.
- Crafted a recommendation tool with **100% accuracy** in filtering sensitive content, ensuring **zero exposure** for at-risk users.
- Optimized algorithms with **collaborative filtering** and custom weights based on **5 user profiles** and **3 content categories** for balanced recommendations.
- Presented findings at the **Sigma Xi conference** to **20+** participating groups.
- *Tools: Python, NumPy, Pandas, Scikit-learn, Matplotlib, MySQL, NLTK, Microsoft PowerPoint*

PROJECTS

CCTV based Home Security Software | *Python, TensorFlow, Pandas, Numpy, Mediapipe, OpenCV, RESTful API*

- Developed a **real-time CCTV security system using machine learning and computer vision**, integrating OpenCV, Mediapipe, and Face-Recognition API to track and register faces and actions in a CSV database, with a **95%-accurate** neural network to detect suspicious behavior and enhance home security.

AR Cooking Application for Snap Spectacles | *len's Studio, typescript, OpenAI, LLMs*

- Achieved **2nd place** at HackPSU Snap AR Challenge by building a recipe-generation app integrating Snap **Lens Studio** assets and **LLMs** to deliver real-time, interactive cooking instructions, **streamlining** the user experience.

Nutrition Tracking Web App | *Javascript, React.js, Node.js, Express.js, mySQL, Python*

- Built a **full-stack** Nutrition Web App using ReactJS, NodeJS, MySQL, and ExpressJS to track nutrition, offering features like calorie tracking, fitness suggestions, and **CRUD** operations, with a Python script to populate test data for system validation.

ACHIEVEMENTS

- **President, ACM:** Directed a **15-member** team to organize **28+ workshops** and **3 collaborative** events per semester, training students for ICPC contests and overseeing logistics for smooth, impactful events.
- **Committee Member, IEEE Robotics:** Facilitated a Robotics Day fair for K-12 students, **promoting** robotics and **mentoring** young minds in STEM.